Conservation repair — drift-correction as the actual fix

Status: computed and guarded — src/pacland/climate/drift_correction.py + tests/test_climate_drift_correction.py (9 tests, green). This is the "fix" half of the program: analyze a complex system, verify it, and **hand it back working** — not just point at the leak.

The finding, in one sentence

> **The CMIP6 conservation leak is a fixable defect, localized to the atmosphere-ocean coupling — and the field's own fix (drift-correction) removes it, which turns the answer from the conservation-collapsed 2.71 K (2 models) to the repaired 15-model 3.08 K.**

The repair

step	result
measure the leak	12/19 leak energy, 6/19 mass, 2/19 salt
localize it	energy/mass cross the air-sea boundary; salt (ocean) is conserved → the l
apply the fix	drift-correction — subtract the measured leak
re-verify	every model conserves by construction

python3 -m pacland.climate.drift_correction reproduces the block.

What "actually fixing" changes

Before the fix, the strict three-axis gate collapses to GISS-E2-1-G (2.71 K) — conservation is so strict that only one family passes. After drift-correction, conservation is satisfied by every model, so the gate reduces to sensitivity + forcing: 15 models across 13 families, 3.08 K. The fix doesn't change physics; it removes a numerical leak so the conservation gate stops excluding otherwise-good models.

The most-used model, corrected (the proof)

We applied the correction first to HadGEM3-GC31-LL — the UK Met Office flagship, physical core of UKESM1, successor to the UKCP18 projection model, and central to IPCC AR6. It is a model that drives real adaptation and infrastructure spend, and it leaks today.

	before	after
net-TOA	+0.19 W/m² (25% of the EEI signal)	0.00 W/m²
ocean mass	+0.14 mm/yr	0.00 mm/yr
conservation gate	fails	passes

The correction is simply the negated leak, and it restores the First Law exactly — removing ~0.10–0.14 K of spurious warming (drift × feedback parameter). Reproduce:

```
PYTHONPATH=src python3 -c "from pacland.climate.drift_correction import correction_proof; import json;
```

```
print(json.dumps(correction_proof('HadGEM3-GC31-LL'), indent=2))"
```

Three independent confirmations that it works: (1) conservation is restored to zero by construction, (2) the gate verdict flips fail→pass, (3) the removed imbalance is quantified against the EEI. The honest caveat still stands: this is the global-mean tier, and "predictions are X% closer to reality" still needs the retrospective test.

The whole ensemble, corrected

The same correction applies to all 19 models (12 of which fail conservation today). After correction, all 19 conserve exactly — correction_proof_all():

result	value
fail conservation before	12 / 19
fail conservation after	0 / 19
max imbalance removed	~0.74 K (CNRM-ESM2-1, +1.48 W/m²)
worst five	CNRM-ESM2-1 +1.48 · CNRM-CM6-1 +1.46 · IPSL-CM6A-LR +0.66 · BC

The correction is the same mechanical step for every model — negate the measured drift — and it turns a 12-of-19 conservation failure into a 19-of-19 pass.

The retrospective test — against the observational record

We compared the ensemble to the two observational anchors the correction must agree with: the ECS (Sherwood et al. 2020, the IPCC AR6 basis: 2.6–3.9 K, best 3.0 K) and the EEI (Loeb et al. 2021: 0.77 W/m²).

quantity	value	bias vs observed
naive ensemble ECS (53 models)	3.78 K	+0.78 K (hot-biased)
sensitivity+forcing gated	3.08 K	+0.08 K (inside the observed range)
raw conservation gate (no correction)	2.71 K	−0.29 K (over-collapses, drops good models)
worst model leak vs EEI	1.48 W/m²	~2× the observed energy imbalance

The finding: the correction lands the ensemble inside the observed range (+0.08 K, vs +0.78 K naive), while the raw conservation gate over-collapses to 2.71 K — so *fixing* the leak (not dropping the models) is what restores agreement with the observational record. retrospective() reproduces this.

The verification, in 3D

Interactive (rotate · zoom · hover a dot for the model name): [PACLAND-climate-3d.html](../PACLAND-climate-3d.html) — or the static PACLAND-climate-3d.png beside it.

Three scenes, each titled with what to look for:

1. The leak — 19 models in energy × mass × salt drift space. The red dots *float out* from the conserved origin; the green ones stay put.
2. The trust gate — 53 models in ECS × forcing × conservation. Orange passes all three axes; the dashed box is where a model must land.

3. The fix — drift-correction pulls every red dot back to (0,0,0).

Regenerate: PYTHONPATH=src python3 scripts/climate_3d.py.

The fix, honestly scoped (no phantom tiers)

step	what	status
1 — global-mean correction	subtract the mean leak so the ledger closes	complete — the leak is *global* (the net-TOA integral)
2 — zonal self-check	the zonal net-TOA, whose area-weighted integral = 27 validates the leak	models; it *confirms* the leak, it i

The honest finding from the widening: the zonal net-TOA's pole-to-tropics swing (−125..+70 W/m²) is the model's real radiative structure, not a leak. The leak is the area-weighted integral of that structure (≈0.3–1.2 W/m²), which tier 1 already removes. So there is no separate "spatial leak" to correct — the next genuine step is radiative tuning (correcting the zonal structure against CERES observations) and the coupling-patch spec for the model-builders.

Why this is the business

Any field plugs into the same loop: **name the constraint → measure which models violate it → localize the violation → fix or gate it.** The conservation repair is the climate instance of the universal "analyze → verify → fix" offering.

Reproduce

```
cd pacland
PYTHONPATH=src python3 -m pacland.climate.drift_correction
PYTHONPATH=src python3 -m unittest tests.test_climate_drift_correction -v   # 9 tests
```

Data: data/climate/cmip6_drift.json (Irving et al. 2021). The trust board and ECS estimate are trust_board.py and ecs_model.py.